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Machine Learning II  
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## Machine Learning II Final Guide

This repository contains two machine learning projects:

Regression: Predict building heating load using architectural features.

Classification: Predict loan default risk using financial and demographic data.

Both notebooks are fully reproducible and include data preprocessing, model tuning, evaluation, and visualizations.

### **Files:**

- `Regression.csv` – Energy efficiency dataset (768 samples)
- `Classification.csv` – Synthetic loan approval dataset (45,000 samples)
- `Regression.ipynb` – Jupyter notebook for regression task
- `Classification.ipynb` – Jupyter notebook for classification task

### **Instructions:**

1. Place all files in the same folder.
2. Download Classification and Regression csv files.
3. Open either `.ipynb` file in Jupyter Notebook.
4. Upload the respective Csv file for Classification and Regression.
5. Run all cells sequentially.

### **Tech Stack:**

- Python 3
- pandas, NumPy
- scikit-learn
- XGBoost
- Matplotlib, Seaborn
- Jupyter Notebook / Google Colab

## **Reproducibility:**

All results can be reproduced by running the notebooks from top to bottom.  
Random seeds are fixed where applicable to ensure consistent results.

## **Results:**

### **Classification: Loan Default Prediction**

#### **Best Model: XGBoost Classifier**

- AUC: 0.975  
The model can distinguish between borrowers who will default and those who won't with a 0.975 confidence rating.
  - Accuracy: 92.9%  
The model correctly classifies over 92% of all loan applicants.
  - Precision (default class): 89%  
When the model predicts a loan will default, it is correct 89% of the time.
  - Recall (default class): 77%  
The model identifies 77% of all actual default cases.
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### **Regression: Building Heating Load Prediction**

#### **Best Model: Gradient Boosting Regressor**

- RMSE: 0.524 kWh  
On average, the model's predictions are off by about 0.52 kilowatt-hours from the true heating load.
- MAE: 0.386 kWh  
The average absolute error is 0.39 kWh, meaning most predictions are within less than half a kWh of the actual value.
- $R^2$ : 0.997  
The model explains 99.7% of the variability in heating load based on building features.

## **Conclusion:**

- Gradient Boosting in regressions helps predict building energy use accurately, supporting greener, smarter design.
- XGBoost in classifications makes loan decisions fairer and more reliable—even when data is imbalanced.

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